StandAlone Carton / Pallet / EOL Label System

Plant-Floor Architecture

Daltile plant-floor label printing pipeline — one C# application (`StandAlone.CartonUi.exe`), deployed to three physical stations, bridging PLC/sorter hardware and thermal printers to enterprise systems: item master data sourced from the Progress OpenEdge database (MDM), and production backflush confirmations sent to SAP.

- External system
- PLC / scanner hardware
- CSV / file
- Application logic
- Thermal printer
- Shared network resource
- SAP backflush integration

EXTERNAL SYSTEM · MASTER DATA

Progress OpenEdge Database — Item Master (MDM)

`itemdet` + `itemhdr` tables (schema `bcmstr3.df`). This is the system of record for item numbers, descriptions, UPCs, shade/size and pallet packing rules.

EXTERNAL SYSTEM · ERP

SAP ERP

Receives production-confirmation / **backflush** transactions once a pallet is confirmed at End-of-Line.

scheduled export

SHARED INPUT FILE · REFRESHED PERIODICALLY

Item Master Export

`itemdet.csv` (US items) & `mitemdet.csv` (Mexico items) — 43-column exports dropped into the shared `data\` directory. Read by both the Carton and Pallet stations for every label decision.

STATION 1

Carton-Label Printing

`MainForm · no launch flag`

PLC-driven: prints a carton label as each carton is sorted/stacked.

STATION 2

Pallet-Label Printing

`MainForm (manual) & --pallet`

Operator-driven: scans a carton, prints the pallet label it belongs on.

STATION 3

EOL Scanning

`--eol → EolScanForm`

Operator-driven: confirms a finished pallet leaving the line and backflushes SAP.

read: itemdet.csv

read: itemdet.csv

PLC / SORTER HARDWARE

Nuovofina Sorter PLC

Serial or TCP feed. Each 65-char frame carries stack #, item

OPERATOR INPUT

Scan / type carton barcode

Operator scans the printed carton barcode (or enters item #/stack #) to identify what's on the pallet.

OPERATOR INPUT

Scan pallet serial

Operator scans the pallet's printed serial as it leaves the line (accepts dashed or bare 12-digit form).


#, and a print-trigger
writes
looks up
B
PRODUCED + READ HERE
boxes{NN}.csv & stackers{NN}.csv
One row per carton scanned (with computed barcode), plus current stack→item configuration. Appended continuously by the PLC handler; read back fo
rowse
grid and
prints.
reads
LOOKUP
Match against boxes{NN}.csv
Resolves the scanned carton to its stored barcode, item, and stack — falls back to item# or stack# match if barcode isn't found.
match
validates
APPLICATION LOGIC
Label decision + SATO template engine
Resolves a .sato template by label size/type, renders carton barcode + item data, converts to raw
commands.
prints
barcode written back
APPLICATION LOGIC
Pallet Query / PalletScanForm
the next pallet serial (pallet-serial-{plant}.txt) and builds the WMS 4×6 pallet label payload.
prints
APPLICATION LOGIC
EolScanForm confirmation
Validates the scanned serial exists in the pallet registry, then logs the scan and builds the SAP confirmation payload.
logs scan
sends
THERMAL PRINTER
Carton Printer
SATO / ZPL / IPL — prints the carton barcode label (barcode, item info, UPC). Barcode is written back to boxes{NN}.csv so reprints stay consistent.
THERMAL PRINTER
Pallet Printer
SATO WMS 4×6 pallet label — Tag / SKU / Shade / Plant / Qty / Grade / Shift / Line, with the source carton barcode embedded as a reference.
writes
shared read
LOCAL AUDIT LOG
eol-scans.csv
Every EOL confirmation logged locally — survives restarts, backs the "last 15 scans" grid.
OUTBOUND INTEGRATION FILE
sap-output/*.xml
One backflush confirmation XML per EOL scan — pallet-level qty (no operation/route split).

SHARED NETWORK RESOURCE

`pallets.csv` — **Pallet Registry**

Written by **every** successful pallet print (Station 2). Must point to the *same* network path (`AppSettings.PalletsCsvPath`) from every station, or EOL can only confirm pallets printed on its own PC.

Deployment note — all four modes are the **same** executable, launched with a different command-line flag (`(none)` / `--pallet` / `--eol` / `--settings`), each with its own local `carton-settings.json` (PLC address, printer address, CSV directory, production line #). Item master files (`itemdet.csv` / `mitemdet.csv`) are refreshed from the Progress/MDM export on a scheduled basis, not real-time. A separate, disconnected prototype pipeline (`StandAlone.Console` / `StandAlone.Worker`) exists in the same solution but is not part of this production flow.